Robert Grou-Szabo

Electrical Engineer — Hardware, Firmware, VLSI / Computer Vision

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Profile

Electrical engineer specializing in **hardware and firmware**, with strong experience in **ASIC/FPGA** design and **VLSI** algorithms for **computer vision** and **video processing**. Career path from graphic design and 3D animation interests through doctoral work at the University of Tokyo. Values tangible hardware outcomes, continuous learning, and applied AI. Holds a valid **European (EU) passport**, enabling residence and work in the EU/EEA without a work visa and free movement in the Schengen area—practical for on-site client visits and assignments in Europe.

Professional Experience

Tamaggo — Montreal, Canada **Mar 2013 – May 2014**
Firmware / Embedded C++ Engineer
Embedded development focused on image-processing products.

- Firmware and embedded software in C++ (and assembler) for microcontrollers and on-board processors.
- Video processing algorithms, noise filtering, and real-time image processing.
- Product features involving video (and related audio) in real-time embedded environments.

Star Systems — Tokyo, Japan **Mar 2010 – Jun 2010**
Embedded Systems Design (contract)
Next-generation motion sensor for the Axbo sleep-tracking alarm clock.

- Low-power sensor module based on Atmel AVR (assembler), with tight memory and power constraints.
- Integration of dual 2-axis tilt sensors and BlueGiga WT12 Bluetooth (I^2C).
- USB charging and firmware update path for Li-Ion battery.

Toin Corporation — Tokyo, Japan **Dec 2008 – Apr 2010**
Japanese–English Translator / Proof-reader (contract)

- Linguistic and contextual QA of JP–EN translations for PC, consoles, PSP, Nintendo DS, and mobile applications.

The University of Tokyo — Tokyo, Japan **Apr 2005 – Mar 2006**
Research Associate

- Video noise detection and reduction algorithms.
- Matlab and C++ work on color correction, histogram equalization, JPEG quantization noise, motion blur, edge/corner detection, optical flow, Hough transform, and template matching.

Gennum Corporation — Burlington, Ontario, Canada **Jun 2003 – Dec 2005**
Software Programmer
Reconfigurable Kalman video noise filter targeting HDTV rates.

- Configurable data-path width, pipeline depth, and buffer sizes; reusable noise-filter library.
- Full flow: Matlab simulation, SystemC validation, VHDL implementation, Xilinx Virtex FPGA bring-up, C# GUI.

École Polytechnique de Montréal — Montreal, Canada **Mar 2001 – Sep 2002**
Research Associate

- Support for doctoral research: C++, assembler, VHDL; hardware/software co-design, test, and simulation.

IBM — Essex Junction, Vermont, USA **Jun 2000 – Mar 2001**
GUI Designer / Software Programmer

- C++ (MFC) application to convert and edit design files between Synopsys formats and IBM proprietary formats (Tcl/Tk, VHDL, Verilog, Primetime, Einstimer).

Metronome Productions Inc. — Montreal, Canada **Sep 1995 – May 1996**
3D Graphic Designer / Modeller (contract)

- 3D models and renders for broadcast commercials (LightWave 3D, 3D Studio MAX); video and audio post (Premiere, Pro Tools, Sound Forge); blue-screen compositing.

Education

Ph.D., Electrical Engineering (Microelectronics) **Mar 2011**
The University of Tokyo, Japan
Thesis: *Automatic Image Noise Type Determination Based on Directional Edge Information*

M.Sc.A., Electrical Engineering (Microelectronics) **Dec 2005**
École Polytechnique, Université de Montréal, Canada
Thesis: *Reconfigurable Platform for Streaming Data Flow Applications*

B.Eng., Electrical Engineering (Telecommunications) **Dec 2000**
École Polytechnique, Université de Montréal, Canada

B.Eng., Electrical Engineering **Jun 1997**
École de technologie supérieure (ÉTS), Université du Québec à Montréal, Canada

DEC, Electronics (Computer Hardware Design) **May 1994**
Cégep de Saint-Laurent, Montreal, Canada

Awards, Grants & Scholarships

- Research Associate Grant — École Polytechnique
- Scholarship Grant — Gennum
- Monbukagakusho Scholarship — Japanese Ministry of Education
- Research Associate Grant — Global Center of Excellence

Selected Publications

Peer-reviewed conference and journal papers. Additional posters during graduate studies are omitted.
IEEE author profile: https://ieeexplore.ieee.org/author/38272216600

1. R. Grou-Szabo, T. Shibata, "A Dominant-Noise Discrimination System for Images Corrupted by Content-Independent Noises Without a priori References," IEEE ICSPCS 2011. IEEE Xplore
2. R. Grou-Szabo, T. Shibata, "Blind Motion-Blur Parameter Estimation Using Edge Detectors," IEEE ICSPCS 2009. IEEE Xplore
3. R. Grou-Szabo, T. Shibata, "Random-Valued Impulse Noise Detector for Switching Median Filters Using Edge Detectors," IEEE ICSPCS 2009. IEEE Xplore
4. R. Grou-Szabo, T. Shibata, "Blind Image Compression History Determination Using Dynamic Thresholding," IEEE ICASSP 2008. IEEE Xplore
5. R. Grou-Szabo, H. Ghattas, Y. Savaria, G. Nicolescu, "Component-based Methodology for Hardware Design of a Dataflow Processing Network," IEEE IWSOC 2005. IEEE Xplore
6. E. Granger, S. Catudal, R. Grou-Szabo, M. M. Mbaye, Y. Savaria, "On Current Strategies for Hardware Acceleration of Digital Image Restoration Filters," *WSEAS Transactions on Electronics*, 2004.
7. P. Nsame, R. Grou-Szabo, Y. Savaria, "INTIME: a multi-tool specification environment for ensuring timing constraints integrity for SOC design," IP Based Design 2000, Grenoble, France.

Technical Skills

- **Languages:** C/C++/C# (MFC, .NET, OpenGL, SystemC); assembler (Intel, Motorola, Atmel AVR); Python; Matlab/Simulink; VHDL/SystemVerilog; Java (Android); HTML
- **Hardware & embedded:** Xilinx and Altera/Intel FPGA; AVR, 80x, 68k, Arduino; Raspberry Pi; HW/SW co-design; low-power design; Bluetooth, I^2C, USB; embedded Linux (Yocto, Buildroot); FreeRTOS / Zephyr; ARM Cortex-M/A toolchains
- **Vision & media:** OpenCV; FFmpeg; GStreamer
- **Tools:** VS Code, Eclipse, Visual Studio, Atmel Studio; Git; Linux (heavy use), Windows, Unix, VxWorks; Synopsys, Mentor, Cadence, OrCAD; SystemC; Docker; CMake; Jupyter
- **Creative / media:** Adobe Creative Suite; 3D Studio MAX, LightWave 3D; Pro Tools, Sound Forge; broadcast production experience
- **Currently learning (personal projects):** PyTorch, TensorFlow/Keras, CUDA

Languages & Mobility

- **English** — native; **French** — native; **Japanese** — intermediate (ongoing study); Hungarian — conversational understanding; Mandarin Chinese — beginner
- Driver's licence and motorcycle licence
- Valid **EU passport** — live/work in EU/EEA without work visa; Schengen mobility for on-site client work

Selected Personal Projects

- **Local home AI / server** — GPU machine running Ollama and related tools; self-hosted services and Git; early Alexa-to-local-Ollama voice experiments.
- **ChronoVault** — local chronological media archive from scattered drives, NAS, and cloud sources, with metadata/OCR-assisted dating. `https://github.com/chronomicron/ChronoVault`
- **VLC scheduled player** — API-driven VLC control with a local show database for fixed daily TV schedules. `https://github.com/chronomicron/VLC_scheduler`
- **IMU orientation** — experimental fusion of gyroscope and accelerometer for PCB orientation estimation.
- **Electronics repair** — ongoing repair/restoration of consoles, computers, TVs, radios (e-waste reduction).
- Game mods (Valheim, RimWorld).

Additional Interests

Self-sufficiency and practical skills (home electrical, plumbing, carpentry, small engines); fermentation and cooking; beekeeping; gardening and bonsai; hiking, cycling, snowboarding; astronomy; tabletop RPGs; long-standing involvement with computing since the mid-1980s (Apple IIe, C64, Amiga, and onward).

Profile site: chronomicron.github.io • Domain: chronomicron.com